

Exercises

Exercise (1)

For an *npn* BJT operating in the active region, find the following:

- (a) The change in the base-emitter voltage v_{BE} if the current i_C is doubled.
- (b) The change in the base-emitter voltage v_{BE} if the current i_C is increased by a factor of 10.
- (c) The percentage change in i_C corresponding to changes in v_{BE} of +0.5 mV, -0.5 mV, +1 mV, -1 mV, +2 mV, -2 mV, +5 mV, -5 mV, +10 mV, and -10 mV. Present the results in a table and comment.
- (d) The values of α and β if at $i_C = 1$ mA the base current i_B is measured as 12.5 μ A. What is the corresponding value of emitter current?
- (e) The saturation current I_S if at $i_C = 1$ mA the base-emitter voltage $v_{BE} = 675$ mV.
- (f) The base-emitter voltage that results if two identical transistors are connected in parallel and the total collector current in the parallel combination is 1 mA.

(a)

$$i_C = I_S e^{v_{BE}/V_T}$$

At $i_C = i_{C1}$,

$$i_{C1} = I_S e^{v_{BE1}/V_T} \quad (1)$$

and at $i_C = i_{C2}$,

$$i_{C2} = I_S e^{v_{BE2}/V_T} \quad (2)$$

Dividing Eq. (2) by Eq. (1), we obtain

$$\frac{i_{C2}}{i_{C1}} = e^{(v_{BE2} - v_{BE1})/V_T} \quad (3)$$

Which can be expressed alternately as

$$v_{BE2} - v_{BE1} = V_T \ln \left(\frac{i_{C2}}{i_{C1}} \right) \quad (4)$$

For $i_{C2}/i_{C1} = 2$,

$$\Delta v_{BE} \equiv v_{BE2} - v_{BE1} = 25 \ln 2 = 17.3 \text{ mV}$$

(b) Using Eq. (4) with $i_{C2}/i_{C1} = 10$, we find

$$\Delta v_{BE} \equiv v_{BE2} - v_{BE1} = 25 \ln(10) = 57.6 \text{ mV}$$

(c) The percentage change in i_C ,

$$\frac{i_{C2} - i_{C1}}{i_{C1}} \times 100$$

corresponding to a change $\Delta v_{BE} \equiv v_{BE2} - v_{BE1}$ can be calculated using Eq. (3). The results obtained for the given values of Δv_{BE} are as in the table below:

We observe that for small Δv_{BE} , positive and negative increments yield nearly equal positive and negative changes in i_C . Also, for small Δv_{BE} , the changes in i_C are linearly related to Δv_{BE} with a proportionality factor of 4% per mV. This linear relationship breaks down when Δv_{BE} becomes large, for instance, ± 10 mV.

$\Delta v_{BE}, \text{mV}$	+0.5	-0.5	+1.0	-1.0	+2.0	-2.0	+5.0	-5.0	+10	-10
$\frac{\Delta i_C}{i_C}, \%$	+2	-2	-4	-4	+8.3	-7.7	+22	-18	+49	-33

(d)

$$\beta = \frac{i_C}{i_B} = \frac{1 \text{ mA}}{12.5 \text{ } \mu\text{A}} = \frac{1}{0.0125} = 80$$

$$\alpha = \frac{\beta}{\beta + 1} = \frac{80}{81} = 0.988$$

$$\begin{aligned} i_E &= i_C + i_B \\ &= 1 + 0.0125 = 1.0125 \text{ mA} \end{aligned}$$

(e)

$$\begin{aligned} 1 \times 10^{-3} &= I_S e^{675/25} \\ \Rightarrow I_S &= 1.88 \times 10^{-15} \text{ A} \end{aligned}$$

(f) With two identical transistors connected in parallel with a combined collector current of 1 mA, each transistor has a collector current of 0.5 mA. Thus, Eq. (4) can be used to determine the change in v_{BE} as i_C decreases from 1 mA to 0.5 mA as

$$v_{BE2} - v_{BE1} = 25 \ln(0.5) = -17.3 \text{ mV}$$

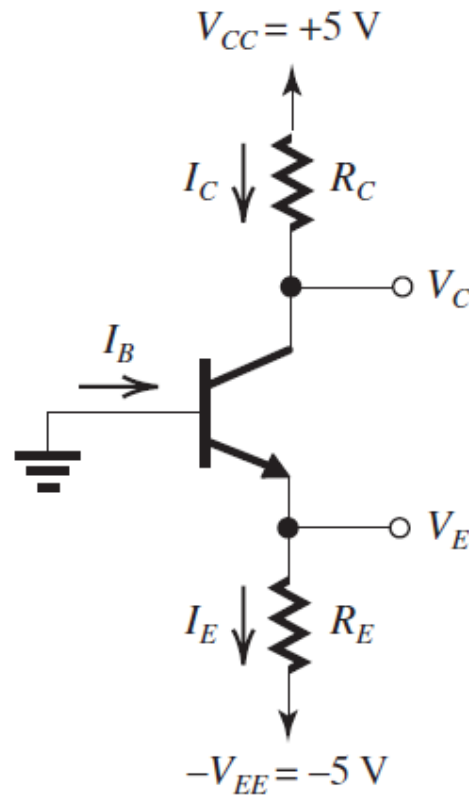
From (e) above, $v_{BE1} = 675 \text{ mV}$, thus

$$v_{BE2} = 675 - 17.3 = 657.7 \text{ mV}$$

Alternatively, we can think of the combination of the two parallel devices as a transistor with twice the base-emitter area and correspondingly twice the value of I_S . When this equivalent transistor is conducting a 1-mA collector current, the base emitter voltage can be obtained from

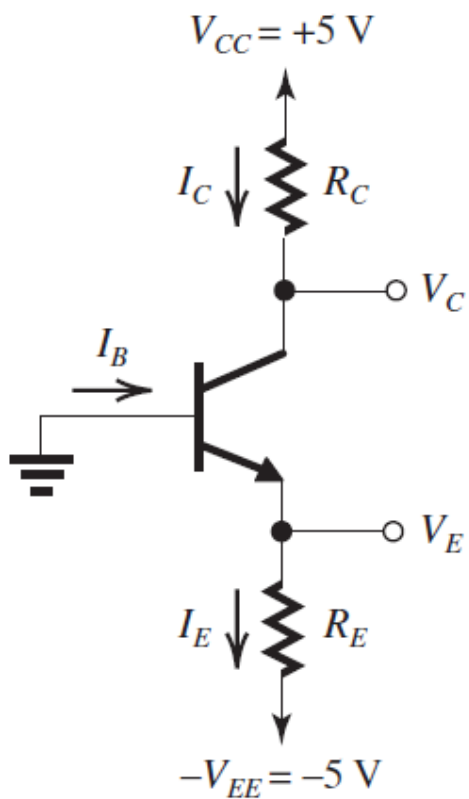
$$\begin{aligned} 1 \times 10^{-3} &= 2 \times 1.88 \times 10^{-15} e^{v_{BE}/V_T} \\ \Rightarrow v_{BE} &= 657.7 \text{ mV} \end{aligned}$$

Exercise (2)



The BJT in the circuit in Fig. 6.2.1 has $\beta = 100$. Resistance R_E is $5\text{ k}\Omega$. The voltages V_E and V_C are measured and found to be -0.68 V and $+1.58\text{ V}$, respectively.

- (a) In what mode is the BJT operating?
- (b) Find I_E , I_B , and I_C .
- (c) What must R_C be?
- (d) Redesign the circuit to obtain a collector current of 2 mA and a collector voltage $V_C = +1\text{ V}$. What are the new values of R_E and R_C ?



(a) $V_C = +1.58\text{ V}$ and $V_B = 0\text{ V}$; thus, $V_C > V_B$, indicating that the transistor must be operating in the active mode.

(b)

$$I_E = \frac{V_E - (-V_{EE})}{R_E} = \frac{-0.68 - (-5)}{5}$$

$$I_E = 0.864\text{ mA}$$

$$I_B = \frac{I_E}{\beta + 1} = \frac{0.864}{100 + 1} = 0.0086\text{ mA}$$

$$I_C = I_E - I_B = 0.855\text{ mA}$$

(c)

$$I_C = \frac{V_{CC} - V_C}{R_C}$$

$$0.855 = \frac{5 - 1.58}{R_C}$$

$$\Rightarrow R_C = 4\text{ k}\Omega$$

(d) To obtain $I_C = 2\text{ mA}$, the emitter current I_E must be

$$I_E = \frac{I_C}{\alpha}$$

where

$$\alpha = \frac{\beta}{\beta + 1} = \frac{100}{101} = 0.99$$

Thus,

$$I_E = \frac{2}{0.99} = 2.02 \text{ mA}$$

Next, we determine V_E by first finding V_{BE} . The transistor has $I_C = 0.855 \text{ mA}$ at $V_{BE} = 0.68 \text{ V}$. To obtain $I_C = 2 \text{ mA}$, V_{BE} must be increased by

$$\Delta V_{BE} = V_T \ln \frac{2}{0.855} = 0.02 \text{ V}$$

resulting in

$$V_{BE} = 0.68 + 0.02 = 0.7 \text{ V}$$

and thus

$$V_E = 0 - V_{BE} = -0.7 \text{ V}$$

The required value of R_E can now be determined from

$$R_E = \frac{V_E - (-V_{EE})}{I_E} = \frac{-0.7 - (-5)}{2.02}$$

$$\Rightarrow R_E = 2.13 \text{ k}\Omega$$

The required value of R_C can be determined from

$$R_C = \frac{V_{CC} - V_C}{I_C} = \frac{5 - 1}{2}$$

$$\Rightarrow R_C = 2 \text{ k}\Omega$$

